MILITARY SPECIFICATION

MUELLER HINTON MEDIUM, DEHYDRATED

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dehydrated Mueller Hinton Medium (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

NATIONAL COMMITTEE FOR CLINICAL LABORATORY STANDARDS

NCCLS Standard ASM-2.

(Application for copies should be addressed to the National Committee for Clinical Laboratory Standards, 771 E. Lancaster Avenue, Villanova, PA 19083.)

3. REQUIREMENTS

3.1 Material. Shall be a stable dehydrated powder suitable for use in Kirby Bauer sensitivity testing of microorganisms to antibiotics and chemotherapeutic agents. The powder shall be cream white to pale tan in color. Mueller Hinton medium shall be in compliance with all applicable requirements of the National Committee for Clinical Laboratory Standards (NCCLS ASM-2) and the Bureau of Medical Devices and Diagnostics.

3.1.1 Ingredients. All ingredients entering into the dehydrated medium shall be of the highest grade suitable for bacteriological use.

3.2 Preparation of end item for use. Thirty-eight (38) grams of dehydrated medium shall make 1 liter of Mueller Hinton medium when suspended in purified, distilled, or deionized water. The Mueller Hinton medium, after being prepared according to manufacturer's instructions, shall dissolve fully and have no lumps.

3.3 Performance characteristics.

3.3.1 Susceptibility (Sensitivity) testing. When the Mueller Hinton medium is tested against cultures of Staphylococcus Aureus (ATCC 25923), Escherichia Coli (ATCC 25922) and Pseudomonas Aeruginosa (ATCC 27853) with the appropriate antibiotic or chemotherapeutic agent listed in table I, the following ranges of zone diameter shall be obtained.

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TABLE I. Sensitivity testing.

Antibiotic	Disk Content	Zone Diameter in mm w/S. aureus ATCC 25923	w/E. coli ATCC 25922	w/P. aeruginosa ATCC 27853
Ampicillin	10 mcg	24-35	15-20	
Bacitracin	10 units	17-22		
Carbenicillin	100 mcg		24-29	20-24
Cephalothin	30 mcg	25-37	18-23	
Chloramphenicol	30 mcg	19-26	21-27	
Clindamycin	2 mcg	23-29		
Colistin	10 mcg		11-15	
Erythromycin	15 mcg	22-30	8-14	
Gentamicin	10 mcg	19-27	19-26	16-21
Kanamycin	30 mcg	19-26	17-25	
Methicillin	5 mcg	17-22		
Neomycin	30 mcg	18-26	17-23	
Novobiocin	30 mcg	22-31		
Oleandomycin	15 mcg	19-28		
Penicillin G	10 units	26-37		
Polymyxin B	300 units	7-13	12-16	
Streptomycin	10 mcg	14-22	12-20	
Tobramycin	10 mcg	19-29	18-26	19-25
Tetracycline	30 mcg	19-28	18-25	
Vancomycin	30 mcg	15-19		
CHEMOTHERAPEUTIC AGENT				
Oxolinic Acid	2 mcg	10-13	20-24	
Trimethoprim/Sulfamethoxazole	1.25 mcg/ 23.75 mcg	26-32	26-34	

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.5 Workmanship. Workmanship shall be first class throughout. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE DIVISION

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of medium produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE II. Sampling.

For visual examination	Inspection level	AQL (percent defectives)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 End item testing. Acceptance number shall be zero for all sample sizes.

TABLE III. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Preparation for use	3.1	S-2	5
Susceptibility testing	3.2	S-2	5
Net Contents	(See 3.12 of PPP-C-186)	-	-

1/ Where possible, the same sample shall be used for two (2) or more tests.

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4.4 Examination. The finished medium shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE IV. Classification of defects.

Categories	Defects 1/
Major	
101	Powder not free from caking.
102	Color of powder not homogeneous.
103	Powder not free of foreign matter.
Minor	
201	Exterior of immediate container not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Examination is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests and examinations. Tests and examinations shall be performed to determine compliance with all requirements of this document and all referenced documents.

4.7.1 Susceptibility test. Susceptibility test shall be performed in accordance with the methodology described in the Federal Register, Vol. 37, No. 191, 30 September 1972, for Kirby-Bauer Susceptibility Testing.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

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5.2.1 Unit of issue. One bottle containing 1/4 lb (113.4 Grams) of medium, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Ninety-six (96) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ninety-six (96) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-039-3388	MUELLER HINTON MEDIUM, DEHYDRATED, 1/4 lb (113.4 Gram)

Preparing activity:
DLA-DM

Project No. 6505-1982

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-M-37833(DLA-DM) MUELLER HINTON MEDIUM, DEHYDRATED

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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